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The incidence and prevalence of epilepsy in the United Kingdom 2013–2018: A retrospective cohort study of UK primary care data

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ABSTRACT

Purpose: The aim of this study was to update overall incidence and prevalence calculations for epilepsy of the United Kingdom (UK) and its constituent nations (England, Northern Ireland, Scotland, and Wales).
Methods: We used data from primary care practices contributing to the Clinical Practice Research Datalink (CRPD), based on the electronic health records of 14 million patients, representing approximately 20% of the population. CPRD contains data from two different health record systems: the Vision clinical system (CPRD GOLD database) and the EMIS Web® clinical system (CPRD Aurum database). We calculated incidence and prevalence rates with 95% confidence intervals (CIs). Data were stratified by age, gender, deprivation, country (England, Scotland, Wales and Northern Ireland) and region (England only).
Results: In the UK, the estimated overall point prevalence for epilepsy was 9.37 per 1000 persons / year (95% CI 9.34–9.40) and the overall estimated incidence rate was 42.68 per 100,000 person-years (95% 42.18–43.18) using the CPRD GOLD database. In England, the estimated incidence (37.41 (95% CI 36.96–37.83)) and prevalence (8.85 (95% CI 8.83–8.87)) was lower (combined databases) compared to figures for Scotland (incidence 47.76 (95% CI 46.15–49.42)); prevalence 10.13 (95% CI 10.06–10.20)) (CPRD GOLD only), Wales (incidence 54.84 (95% CI 52.79–56.95); prevalence 11.40 (95% CI 11.31–11.49)) (CPRD GOLD only) and Northern Ireland (incidence 46.18 (95% CI 43.13–49.90); prevalence 12.08 (95% CI 11.93–12.23))(combined databases). Prevalence and incidence were higher in more deprived regions.
Conclusion: The prevalence and incidence of epilepsy in the UK is broadly in line with other high income countries, showing the usual pattern of high incidence in the young and the old, with a nadir in middle age. The prevalence of epilepsy has fallen slightly since 2011. There is significant geographical variation (between countries and between regions), and a suggestion of a relationship between deprivation and epilepsy which needs further investigation.

1. Introduction

The estimated incidence (50–80 per 100,000 person-years) and prevalence rates (5–10 per 1000 person / year) for epilepsy have been found to be relatively consistent across high income countries [1–3]. Nevertheless, changes in demographics and risk factors mean that the epidemiology of epilepsy may change over time. For instance, in countries with an aging population, the increasing prevalence of diseases of old age such as stroke or dementia could increase the prevalence of epilepsy. Conversely, improved General Practitioner (GP) and specialist training, improved guidance for clinicians, better obstetric care and more accurate differential diagnosis could be leading to a reduction in the number of people with epilepsy. Reductions in relevant pathologies (such as mesial temporal sclerosis) have been reported in high income countries [4], with a resultant fall in epilepsy surgeries, a phenomenon that has also been seen in the United Kingdom (UK) [5]. Given how

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important incidence and prevalence rates are for many aspects of health service planning, and that estimates of these parameters for the UK were most recently published over a decade ago, it seemed timely to re-examine the epidemiology of epilepsy and to explore temporal changes in this country (UK) [6]. The methodology used in the 2011 Joint Epilepsy Council paper [6] was essentially the same as the approach used in this study, using the General Practice Research Database (now CPRD GOLD). The diagnostic code lists are similar but not identical and the anti-seizure medication (ASM) lists expanded to include newer medications.

The aim of this study was to update the epidemiology of epilepsy in the UK specifically

- To estimate the incidence and prevalence of epilepsy by year, age group, gender, country (England, Scotland, Wales and Northern Ireland) and region (England).
- To estimate the incidence and prevalence of epilepsy by practice-level deprivation decile, using the appropriate national Index of Multiple Deprivation (IMD) scale.
- To characterise changes in prevalence and incidence rates since 2011.

2. Methods

2.1. Source data

General practitioners (GP) play a key role in the UK health care system, as they are responsible for most primary health care, maintaining national disease registers, making specialist referrals and monitoring long term treatment. The Quality and Outcomes Framework (QOF) requires GPs to keep a register of patients with important long-term conditions including epilepsy [7]. QOF is mainly a tool for UK health policy makers to ensure that agreed treatment standards for the conditions included are met, but it also provides important and high quality statistical information. Different practices use different electronic clinical records systems. Two of the most widely used are EMIS Web® and Vision® [8]. Clinical Practice Research Datalink (CPRD) is a collaboration which collects data from consenting GP practices using these two clinical systems to create two separate databases (CPRD Aurum = EMIS Web®, CPRD GOLD = Vision®). The data recorded in both databases include diagnoses, demographics, prescriptions, clinical events, preventive care, specialist referrals, and, if uploaded by the primary care team, hospital admissions and their major outcomes. While the structure and the content of the CPRD Aurum and CPRD GOLD databases are very similar, they are not identical. Combined, the two databases include data for over 60 million patients since 1987, with 16 million currently contributing patients. This is the best source of epilepsy diagnostic data in the UK representing approximately 20% of the population.

The United Kingdom (UK) has four constituent nations (England, Wales, Scotland, Northern Ireland). Although the overall framework of the National Health Service (NHS) covers the whole of the UK, the system is not completely uniform with different health care bodies governing its constituent nations. The coverage of the two CPRD databases differs regionally and between the constituent nations of the UK, so both CPRD databases, CPRD GOLD and CPRD Aurum were used for this study. The study period for the current analysis was 1st January 2013 to 31st December 2018.

2.2. Case definition

In order to meet our case definition for active epilepsy patients, the available records of individual patients of all ages had to meet two criteria: 1) a diagnosis of Epilepsy on or before 31/12/2018, and 2) use of an ASM on or before 31/12/2018 and after the first recording of an epilepsy diagnosis (to ensure that the indication for the use of ASM was indeed for epilepsy).

The first criterion was met when patients' records included a Read or SNOMED CT code for Epilepsy within their clinical or referral records. Read codes and SNOMED CT codes are alphanumeric codes that are associated with a Read or SNOMED CT term such as epilepsy or asthma [9]. They provide a standard vocabulary for clinicians to record patient findings and procedures, in health and social care. Out of a total of 115, 984 diagnostic Read codes in CPRD GOLD (Vision® practices) and in excess of 1 million SNOMED CT codes in CPRD Aurum (EMIS Web® practices), 253 CPRD Aurum medical codes and 162 CPRD GOLD medical codes were used to select people with epilepsy. These were selected by the authors as the best chance to capture as many people with epilepsy as possible, including those for whom no specific epilepsy type was found in their record. There is no published consensus on exactly which codes should be used for the identification of epilepsy, and so they were determined by the authors. In addition to directly epilepsy-related diagnostic terms themselves, terms intrinsically related to a diagnosis of epilepsy were included in the search criteria. This included terms such as "epilepsy prevents employment", "petit-mal epilepsy", "Pregnancy advice for patients with epilepsy". These terms, along with a confirmed prescription of an ASM were sufficient to conclude the patient had an active diagnosis of epilepsy.

In order to meet the second inclusion criterion, an ASM prescription, as identified by the authors, had to be recorded within 12 months of the recording of the epilepsy diagnosis. The presence of a diagnostic record and a subsequent ASM prescription was concluded as indicative of active epilepsy. The index date for epilepsy was defined as the date of the first ASM record meeting this criterion. Patients with an index date on or before 31/12/2018 were included. Patients were excluded from the study as cases if their follow-up within the CPRD data ended before the beginning of the study period.

The denominator data was defined as the sum of the person years of follow-up of patients who were alive and registered at a CPRD contributing practice during each calendar year of interest.

The full list of Read and SNOMED CT codes considered as indicative of a diagnosis of epilepsy and the codes for the ASMs is provided in Supplement 1, along with full details of the study design.

In England social deprivation was assessed using the 2015 IMD scale, whilst in the other nations of the UK the 2016 Scottish IMD (SIMD) was used for Scotland, the 2014 Welsh IMD (WIMD) for Wales and the 2017 Northern Ireland IMD for Northern Ireland. Patient-level IMD was used where available (England and those eligible for linkage at the patient level), Practice-level IMD was used as a proxy when the patient-level value was not available.

2.3. Incident cases

Patients were defined as incident if their index date as defined above, occurred 12 months or more after the patient's registration date with the practice and during the patient follow-up period, with the assumption therein that the diagnosis was new (and unique), made in the previous twelve months. It is therefore assumed that an individual patient could only ever be included once (n.b. it is unlikely that patients could have been captured twice because they had moved from one practice contributing to the CPRD dataset to another; in the UK the medical notes follow patients to a new practice, so this should only have happened if a patient's notes had got lost during the course of a transfer). The numerator was defined as the total number of patients who met the definition of an incident case of epilepsy during each calendar year of interest. Age was calculated as the difference between the patient year of birth and the calendar year of the index date.

Incidence rates were calculated by dividing the number of incident cases by the calculated person-time at risk for that year. Incidence rates were expressed as the number of newly diagnosed cases per 100,000 person-years. Incidence rates were stratified separately by year, gender, age categories in 5-year age groups, UK country, region/strategic health

authority and 2015 IMD practice-level deciles.

A figure was calculated for each area for each of the six years. These were then averaged to give the mean annual incidence across the six-year period.

2.4. Prevalent cases

All patients meeting the case inclusion criteria were defined as prevalent cases. Patients had to have their index date before the end of patient follow-up. Patient follow-up end was defined as the earliest of that practice last collection date, the patient transfer out date, the CPRD date of death and the end of the study period: 31/12/2018. There was no requirement for prior registration time before index date for prevalent cases.

Point prevalence rates were calculated at the mid-point of each year within the study period. The prevalence rate was calculated as the total number of cases occurring on or before the 01 July of the calendar year of interest (numerator) divided by the total number of unique individuals in the denominator population. Prevalence was expressed as the number of cases affected by the condition per 1000 patients. As with incidence, prevalence rates were stratified separately by year, gender, 5-year age categories, UK country, 2015 IMD practice-level deciles, and region/SHA.

A figure was calculated for each area for the mid-point for each of the 6 years. These were then averaged to give the mean prevalence across the six-year period at the mid-point.

2.5. Statistical analysis

95% confidence intervals (CI) for the incidence estimate were calculated using a Poisson distribution. 95% confidence intervals (CI) for the prevalence estimate were calculated using a binomial distribution. This analysis was undertaken using the April 2020 static version of the CPRD GOLD and CPRD Aurum databases. Population-weighted estimates using the two databases were calculated for the incidence and prevalence rates in England and Northern Ireland. The UK CPRD GOLD incidence and prevalence figure were calculated by applying the individual incidence and prevalence rates to each country's population and adding these together to give a UK figure. This means the different relative contributions to the base data (more samples as a percentage of the country population for some than others) should not skew the prevalence rate.

3. Results

In 2020 the source population used from the CPRD GOLD database included 3,138,215 individuals, registered with 83 GP practices in England, 179 practices in Scotland, 101 practices in Wales and 35 practices in Northern Ireland (accessed 09/11/2020). In the same year the source population from the CPRD Aurum database included 11,802,119 individuals, registered with 1,233 GP practices in England and 11 practices in Northern Ireland (accessed 09/11/2020).

Table 1
Prevalence by country and gender, CPRD GOLD and CPRD Aurum.

	Country	Gender	Source	Denominator	Cases	Prevalence per 1000	95% CI	Population 2019	Prevalent cases
Prevalence	**England**	**Male**	GOLD	4,190,092	38,935	9.29	9.20–9.38	27,827,831	258,581
		Female	GOLD	4,239,448	37,621	8.87	8.76–8.96	28,459,130	252,547
		Total	GOLD	8,429,540	76,556	9.08	9.02–9.15	56,286,961	511,128
Prevalence	**Northern Ireland**	**Male**	GOLD	783,109	9606	12.27	12.02–12.51	932,717	11,441
		Female	GOLD	789,491	9792	12.40	12.16–12.65	960,950	11,919
		Total	GOLD	1,572,600	19,398	12.33	12.16–12.51	1,893,667	23,360
Prevalence	**Scotland**	**Male**	GOLD	3,663,337	37,845	10.33	10.23–10.43	2,663,003	27,511
		Female	GOLD	3,724,009	36,975	9.93	9.83–10.03	2,800,297	27,804
		Total	GOLD	7,387,346	74,820	10.13	10.06–10.20	5,463,300	55,314
Prevalence	**Wales**	**Male**	GOLD	2,617,413	29,847	11.40	11.28–11.53	1,554,678	17,728
		Female	GOLD	2,637,339	30,074	11.40	11.23–11.57	1,598,201	18,225
		Total	GOLD	5,254,752	59,921	11.40	11.31–11.49	3,152,879	35,953
Prevalence	**United Kingdom** [1]	**Male**	GOLD	11,253,951	116,233	9.56	9.53–9.59	32,978,229	315,261
		Female	GOLD	11,390,287	114,462	9.18	9.15–9.23	33,818,578	310,494
		Total	GOLD	22,644,238	230,695	9.37	9.34–9.39	66,796,807	625,755
Prevalence	**England**	**Male**	Aurum	35,701,713	303,897	8.51	8.48–8.54	28,459,130	242,247
		Female	Aurum	35,712,419	326,218	9.13	9.10–9.17	27,827,831	254,196
		Total	Aurum	71,414,132	630,115	8.82	8.80–8.85	56,286,961	496,443
Prevalence	**Northern Ireland**	**Male**	Aurum	174,180	1825	10.48	10.00–10.97	960,950	10,069
		Female	Aurum	176,895	2014	11.39	10.90–11.89	932,717	10,619
		Total	Aurum	351,075	3839	10.92	10.78–11.07	1,893,667	20,688
Prevalence	**England average** [2]	**Male**		39,891,805	342,832	8.59	8.57–8.62	27,827,831	239,041
		Female		39,951,867	363,839	9.11	9.08–9.14	28,459,130	259,263
		Total		79,843,672	706,671	8.85	8.83–8.87	56,286,961	498,140
Prevalence	**Northern Ireland average** [2]	**Male**		957,289	11,431	11.94	11.72–12.16	932,717	11,137
		Female		966,386	11,806	12.22	12.00–12.44	960,950	11,743
		Total		1,923,675	23,237	12.08	11.93–12.23	1,893,667	22,875

[1] average weighted by country population.
[2] average weighted by sample size of CPRD GOLD and CPRD Aurum.

3.1. Prevalence

The overall estimated prevalence rate (Table 1) for epilepsy in the UK using the CPRD GOLD database was 9.37 per 1,000 persons/year (95% CI 9.34–9.37). In England the prevalence was 9.08 per 1,000 (95% CI 9.02–9.15), in Scotland 10.13 per 1,000 (95% CI 10.06–10.20), in Wales 11.40 per 1,000 (95% CI 11.31–11.49) and in Northern Ireland 12.33 per 1,000 (95% CI 12.16 –12.51). Prevalence estimates for epilepsy in England and Northern Ireland using the CPRD Aurum database were slightly lower, with an estimated prevalence of 8.82 per 1,000 (95% CI 8.80–8.85) in England and 10.92 per 1,000 (95% CI 10.78–11.07) in Northern Ireland. The overall prevalence by country and gender are shown in e-Table 1. The combined estimate for the prevalence of epilepsy in England was 8.85 (95% CI 8.83–8.87) and the prevalence in Northern Ireland was 12.08 (95% CI 11.93–12.23).

Prevalence by country and age group is shown in e-Table 2 and e-Charts 1, 2, 3, 4 and 5 with a trend of increasing prevalence for each 5-year grouping seen in each country until at least the 75–79 age group. Prevalence data by age groups for Northern Ireland using the CPRD Aurum database was not presented due to small numbers and resultant wide confidence intervals,

Prevalence by English regions (using the CPRD Aurum database) is reported in e-Table 3 and e-Chart 6 with the lowest reported prevalence being in London (6.96 per 1,000; 95% CI 6.92–7.01) and the highest prevalence rate being in the North East (11.03 per 1,000; 10.90–11.17).

Prevalence by practice level IMD decile is reported in e-Table 4 and e-Charts 7 and 8, with a clear trend of increasing prevalence with successive levels of deprivation, increasing from a prevalence of 7.86 per 1,000 (95% CI 7.74–7.99) in the lowest IMD decile (1) to a prevalence of 12.07 per 1,000 (95% CI 11.93–12.20) in the highest IMD decile (10 – most deprived) in the CPRD GOLD database (all UK) with a similar picture seen in the CPRD Aurum database (England and Northern Ireland only).

3.2. Incidence

The calculated incidence rates by country and gender are shown in Table 2. The overall estimated incidence for the UK using the CPRD GOLD database was 42.68 per 100,000 person-years (95% CI 42.18–43.18), the estimated incidence rate for England was 41.41 (95% CI 39.99–42.87), for Scotland 47.76 (95% CI 46.15–49.42), for Wales 54.84 per 100,000 (95% CI 52.79–56.95) and for Northern Ireland 45.48 (95% CI 42.13–49.01).

As with the prevalence rates, incidence rates in England and Northern Ireland using the CPRD Aurum database were slightly lower with an estimated incidence rate of 36.93 per 100,000 person-years (95% CI 36.46–37.39) in England and 49.38 (95% CI 42.14–57.52) in Northern Ireland. The combined estimates for the incidence of epilepsy in England was 37.41 (95% CI 36.93–37.83) and the incidence in Northern Ireland was 46.18 (95% CI 43.13–49.90) respectively.

Incidence by country by age group is reported in e-Table 6 and e-Charts 9, 10, 11, 12 and 13, with a pattern of a high incidence rate in the early age groups (0–4, 5–9), with the incidence steadily increasing from the 60–64 age group upwards, with the highest incidence rate seen in the

Table 2
Incidence by country and gender, CPRD GOLD and CPRD Aurum.

				Person-time (years)	Cases	Incidence per 100,000	95% CI	Population	Incident cases
Incidence	**England**	**Male**	GOLD	3,863,067	1610	41.68	39.67–43.76	27,827,831	11,599
		Female	GOLD	3,881,940	1597	41.14	39.15–43.21	28,459,130	11,708
		Total	GOLD	7,745,007	3207	41.41	39.99–42.87	56,286,961	23,307
Incidence	**Northern Ireland**	**Male**	GOLD	750,961	361	48.07	43.24–53.30	932,717	448
		Female	GOLD	755,359	324	42.89	38.35–47.83	960,950	412
		Total	GOLD	1,506,320	685	45.48	42.13–49.01	1,893,667	861
Incidence	**Scotland**	**Male**	GOLD	3,425,722	1615	47.14	44.87–49.50	2,663,003	1255
		Female	GOLD	3,472,923	1680	48.37	46.09–50.74	2,800,297	1355
		Total	GOLD	6,898,645	3295	47.76	46.15–49.42	5,463,300	2609
Incidence	**Wales**	**Male**	GOLD	2,446,551	1298	53.05	50.21–56.02	1,554,678	825
		Female	GOLD	2,452,986	1389	56.62	53.69–59.68	1,598,201	905
		Total	GOLD	4,899,537	2687	54.84	52.79–56.95	3,152,879	1729
Incidence	**United Kingdom** [1]	**Male**	GOLD	10,486,301	4884	42.84	42.13–43.55	32,978,229	14,127
		Female	GOLD	10,563,208	4990	42.52	41.83–43.22	33,818,578	14,380
		Total	GOLD			42.68	42.18–43.18	66,796,807	28,507
Incidence	**England**	**Male**	Aurum	32,617,006	12,244	37.54	36.88–38.21	27,827,831	10,447
		Female	Aurum	32,333,892	11,742	36.31	35.66–36.98	28,459,130	10,334
		Total	Aurum	64,950,898	23,986	36.93	36.46–36.93	56,286,961	20,786
Incidence	**Northern Ireland**	**Male**	Aurum	168,471	74	43.92	34.49–55.14	932,717	410
		Female	Aurum	165,643	91	54.94	44.23–67.45	960,950	528
		Total	Aurum	334,114	165	49.38	42.14–57.52	1,893,667	935
Incidence	**England average** [2]	**Male**		36,480,073	13,854	37.98	37.35 - 38.61	27,827,831	10,569
		Female		36,215,832	13,339	36.83	36.21–37.46	28,459,130	10,481
		Total		72,695,905	27,193	37.41	36.96–37.85	56,286,961	21,057
Incidence	**Northern Ireland average** [2]	**Male**		919,432	435	47.31	42.97–51.97	932,717	441
		Female		921,002	415	45.06	40.83–49.61	960,950	433
		Total		1,840,434	850	46.18	43.13–49.90	1,893,667	874

[1] average weighted by country population.
[2] average weighted by sample size of CPRD GOLD and CPRD Aurum.

85–89 age group in all 4 countries.

Incidence by English region is reported in e-Table 7 and e-Chart 14, mirroring the prevalence figures with the lowest incidence rate being in London 28.5 per 100,000 person-years (95% CI 27.56–29.46) and the highest incidence rate in the North East 41.69 (95% CI 39.07–44.44).

Incidence by practice level IMD decile is reported in Table 3 and e-Table 8 and e-Charts 15 and 16, with the results mirroring that seen for prevalence with incidence rates increasing from the least deprived (IMD decile level 1) 35.6 per 100,000 persons-years (95% CI 32.88–38.48) to the most deprived (IMD decile 10) 58.35 (95% CI 55.32–61.50) in the CPRD GOLD database (all UK) with a similar picture seen in the CPRD Aurum database (England and Northern Ireland). Increasing levels of social deprivation appear to be highly correlated to epilepsy incidence ($r = 0.68$) and prevalence (CPRD GOLD: $r = 0.97$); CPRD Aurum: $r = 0.93$) rates.

4. Discussion

The principal aim of this study was to provide updated estimates for the incidence and prevalence of epilepsy in the United Kingdom (UK). Our analysis revealed an overall estimated prevalence rate of 9.37 per 1000 using the CPRD GOLD database, with slightly lower estimates for the prevalence in England and Northern Ireland respectively. This suggests that the overall prevalence has fallen slightly since 2011. The trends seen in the prevalence figures were mirrored in the incidence figures with an estimated overall incidence of 42.68 per 100,000 person-years using the CPRD GOLD database, with slightly lower estimates (for England and Northern Ireland) using the CPRD Aurum database.

These figures indicate that the reported incidence and prevalence of epilepsy in the UK is broadly in line with that of other high-income countries, with an overall estimate of the annual incidence of epilepsy of 45 per 100,000 (IQR 30.3–66.7) calculated in a systematic review [1], with increasingly higher estimates for Scotland, Wales and Northern Ireland. A more recent meta-analysis of 48 studies suggesting a slightly higher incidence rate of 61.44 per 100,000 person-years (95% CI 50.75–74.38), although not limited to higher income countries [3]. Looking at the age distributions for the incidence of epilepsy, it is clear that epilepsy is increasingly becoming a condition of the elderly, both in terms of a new diagnosis (for example we found the highest age-related incidence rate of 102.16 among those aged 85–90 in England) but also in terms of prevalence, a finding that was replicated in all four constituent nations of the United Kingdom.

The flattening of the expected U-shaped age-related incidence curve is noted, especially at the younger age range. The reasons for this are not clear but could be linked to improved guidance for clinicians, better obstetric care, more effective treatment of febrile seizures and more accurate differential diagnosis in the younger age groups. This merits further study.

Table 3
Incidence by IMD decile, UK, CPRD GOLD.

Source	Practice-level IMD decile	Person-time (years)	Cases	Incidence per 100,000	95% CI
GOLD*	1 (least deprived)	1,778,214	633	35.60	32.88–38.48
	2	1,805,043	769	42.60	39.64–45.72
	3	1,728,763	877	50.73	47.43–54.20
	4	1690,975	652	38.56	35.65–41.63
	5	2,044,958	819	40.05	37.35–42.89
	6	2,277,444	1230	54.01	51.03–57.11
	7	2,721,078	1311	48.18	45.61–50.86
	8	1,874,073	886	47.28	44.21–50.50
	9	2,753,600	1311	47.61	45.07–50.26
	10 (most deprived)	2,375,362	1386	58.35	55.32–61.50

* Covers all four countries.

It is unclear why the incidence was 10% lower in the 90+ age group than in those aged 85–90. There are reports of lower referral rates to regional epilepsy services for people aged 85 and over than younger adults [10]. A number of factors may deter primary care physicians from referring very old, frail, essentially housebound patients to specialists e.g. patient does not want referral, travel is very difficult to hospital clinics. It is also conceivable that milder presentations of epilepsy (such as focal aware seizures or brief focal impaired awareness seizures) are less likely to be recognised as epilepsy or as requiring treatment in the presence of other symptoms of senescence or age-related comorbidities. This observation does not do much to diminish the general trend of a higher prevalence of epilepsy in older age.

Other key findings in this study are the significant regional variations (in England) in the incidence and prevalence of epilepsy, with the highest incidence (and prevalence) seen in the North East, which according to a recent workforce survey by the Association of British Neurologists (ABN), has the lowest concentration of neurologists, whilst the lowest prevalence figures were seen in London, which has the highest concentration of neurologists [11].

There are various potential reasons for differences across English regions and the countries. These could include differences in deprivation, age of population, ethnicities, regional coding practice etc. although the study did not identify any such differences.

Finally, this study suggests a correlation between social deprivation and epilepsy, with the incidence and prevalence in England and Wales increasing from the lowest to the highest IMD decile with an over 40% increase, replicating more broadly a finding that had previously been demonstrated more locally in London [12] and in Wales [13]. This study was neither designed nor does the data allow us to determine whether the possible correlation between the incidence and prevalence of epilepsy and IMD is due to poverty-associated factors directly or variation in healthcare provision and diagnosis (for example lower neurologist ratios in more deprived areas). Interestingly a recent UK study in dementia found that higher socioeconomic deprivation was associated with poorer access to specialist diagnosis (and higher mortality) [14].

We had postulated that the prevalence of epilepsy in the UK may be increasing given the increasing age of the UK population, yet our findings suggest a slight decrease relative to the 2011 estimates [6]. The prevalence for people aged 18 and over in England has been calculated by applying the 15–19 prevalence rate to the 18 and 19 English population and adding it to the calculations for those aged over 20. This gives for people aged 18 and over in England of 10.19 per 1000 in CPRD Aurum and 10.53 per 1000 in CPRD GOLD. This is 21% and 32% higher than the prevalence currently quoted by Public Health England for people aged 18 and which this body with key responsibilities for health service planning uses for decision-making purposes (8 per 1000 for people aged 18 and over in England) [15].

There are likely to be a number of reasons for the difference between the prevalence and incidence figures calculated in this study and those cited by PHE. One possible reason for the higher figures calculated in this study may be the difference in case definition – with PHE defining epilepsy cases as having 'recurrent unprovoked seizures' whereas this study required patients only to be receiving at least one ASM prescription after their diagnosis. The fact that the PHE numbers refer to adults only cannot explain the lower prevalence and incidence figures because these figures are, on average, higher in those under the age of 18 than across the adult age range. The fact that PHE uses Read code data from the whole country might suggest that its figures are more reliable, but extensive epidemiological work with the CPRD database has previously documented that the data it contains is representative of the whole population of the UK [16,17].

To some extent the lower prevalence and incidence rate generated by PHE's analysis of QOF data may be explained by differences in case definition i.e. the Read codes used. NHS Digital (email correspondence) have confirmed that the original code list for including people with epilepsy was based on age criteria and therefore excluded some

"juvenile" forms of epilepsy such as juvenile absence and myoclonic epilepsies amongst other, which is likely to reduce prevalence figures in those above age 17. In particular Juvenile Myoclonic Epilepsy (accounting for 5% - 11% of all epilepsies) and Juvenile Absence Epilepsy (accounting for 2%−3% of the total epilepsy prevalence in adults in previous studies) [18,19] are likely to be particularly relevant omissions. This study highlights the need for a consensus method for defining epilepsy in the UK using primary care data.

The major limitation of this study, and one inherent in all large database studies, is the accuracy of the coding and the inevitable variation seen, a fact seen in the slight different figures estimates given by the CPRD Aurum and CPRD GOLD databases. One of the major concerns with the use of the CRPD database, is that individual practices can opt in or out, representing a potential source of bias. Nevertheless, the CRPD has previously been demonstrated to be representative of the UK population [20–22]. Moreover, we employed wider diagnostic criteria than just epilepsy type to capture those with a current active diagnosis or a new diagnosis of epilepsy. In addition, there was insufficient data available to report prevalence and incidence by English region from CPRD GOLD database. A further limitation of the data relates to the relationship between social deprivation and the incidence and prevalence of epilepsy. Whilst the data suggests a strong relationship, the nature of the data does not allow to establish correlation between the two. In order to accurately establish a correlation would require prospective cohort data to see if social deprivation changed over time in those with a new (incident cases) and pre-existing (prevalent cases) over time.

The diagnostic codes used could conceivably lead to an over-representation of epilepsy diagnosis when codes such as "o/e fit" "o/e - a seizure" might reasonably apply to non-epileptic seizures such as provoked or dissociative seizures. One of these codes might be combined together with a prescription of an anti-seizure medication for an indication such as neuropathic pain or anxiety. While we do not expected this possibility to have a major effect on our findings, future research could greatly benefit from a consensus exercise to produce an agreed list of diagnostic codes to be used in primary care for the diagnosis of epilepsy.

In summary, this study provides updated estimates for the incidence and prevalence of epilepsy in the UK and the constituent nations of the UK, as well as estimates by gender, age group and regional variations. Although our figures indicate that the prevalence of epilepsy has fallen slightly relative to 2011, the estimates generated by our analysis are significantly higher than prevalence figures used by Public Health England for national healthcare planning purposes. Our findings provide further support for the correlation between the incidence (and prevalence) and social deprivation with an almost 40% increase in the incidence (and prevalence) from the lowest to the highest deprivation decile. Finally, as a result of this study and the highlighted variation in coding employed, NHS Digital have confirmed, going forward, that they will include all missing epilepsy codes, allowing for more accurate estimates for the incidence and prevalence of epilepsy.

Author contributions

SW, MR, JD and AN were involved in the study design.

EY and TA undertook the database analysis and produced the initial results and statistical analysis.

SW and AN undertook the analysis and wrote the first draft.

AN undertook additional statistical analysis.

SW, MR, JD and AN were involved in interpretation of data.

All authors were involved in the critical revision of the manuscript.

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Supplement 1 = Incidence and Prevalence of the Epilepsy in the United Kingdom during 2013–2018. CPRD Protocol Number: 19_237. Thamina Anjuman and Eleanor Yelland

Declaration of Competing Interest

The authors have no conflicts of interest to declare

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.seizure.2023.01.003.

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